How to Calculate Rolling Correlation in Python?

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Correlation generally determines the relationship between **two variables**. The rolling **correlation** measure the correlation between **two-time series data** on a rolling window Rolling correlation can be applied to a **specific window width** to determine short-term correlations.

Calculating Rolling Correlation in Python

Let's use **sales data of two products A and B in the last 60 months** to calculate the rolling correlation. Pandas package provides a function called **rolling.corr()** to calculate the rolling correlation.

Syntax:

```
data1.rolling(width).corr(data2)
```

Where,

- **data1, data2 -** data/column of interest (type series)
- **width** - Rolling window width (int)

Note: The **width of the rolling window** should be **3 or greater** in order to calculate correlations.

Data Used:

```python
# import pandas module
import pandas as pd

# read the data
data = pd.read_csv('product_sales.csv')

# display top 10 rows
print(data.head(10))

# display column names
print(data.columns)
```

Output:

Data sample:

```
        Product A  Product B
Months
1              25         30
2              27         31
3              25         33
4              30         34
5              27         35
6              24         30
7              23         28
8              25         30
9              24         33
10             26         36
```

columns:

```
Index(['Product A', 'Product B'], dtype='object')
```

Example 2:

Here, we used the window width of 6, which shows the successive 6 months rolling correlation. We could see a significant correlation between two products sales any sudden dip or rise in correlation signals an unusual event, that caused the dip.

```
data['Product A'].rolling(6).corr(data['Product B'])

# formatting the output
k = 1
for i, j in enumerate(data['Product A'].rolling(6).corr(data['Product B'])):
    if (i >= 5 and i < 12):
        print(f'The correlation in sales during months\
        {k} through {i+1} is {j}')
        i = 0
        k += 1
```

Output:

```
The correlation in sales during months 1 through 6 is 0.6354163283417376
The correlation in sales during months 2 through 7 is 0.7488028723414553
The correlation in sales during months 3 through 8 is 0.7991732049750839
The correlation in sales during months 4 through 9 is 0.7351950265602317
The correlation in sales during months 5 through 10 is 0.8163709564049727
The correlation in sales during months 6 through 11 is 0.5031546054266153
The correlation in sales during months 7 through 12 is 0.5559276045138895
```

Now's let us try the same for 3-month correlation as shown below,

Example 3:

```
data['Product A'].rolling(3).corr(data['Product B'])

# formatting the output
k = 1
for i, j in enumerate(data['Product A'].rolling(3).corr(data['Product B'])):
    if (i >= 3 and i < 12):
        print(
            f'The correlation in sales during months {k} \
            through {i+1} is {j}')
        i = 0
        k += 1
```

Output:

```
The correlation in sales during months 1 through 4 is 0.4335549847620939
The correlation in sales during months 2 through 5 is 0.39735970711955665
The correlation in sales during months 3 through 6 is 0.7559289460184538
The correlation in sales during months 4 through 7 is 0.9992600812897133
The correlation in sales during months 5 through 8 is 0.8660254037843838
The correlation in sales during months 6 through 9 is 0.3973597071194888
The correlation in sales during months 7 through 10 is 0.49999999999999795
The correlation in sales during months 8 through 11 is 0.9607689228305074
The correlation in sales during months 9 through 12 is 0.8386278693775306
```

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